MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 4702: MODELING AND SIMULATION

END OF SEMESTER EXAMINATION

DATE: 10th JULY 2023 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

There are 7 questions in this paper. **ANSWER ALL.**

There are **THREE** pages. **PLEASE CHECK**

Marks for each question are indicated on the right margin.

Write your registration number on top of each answer sheet provided.

Calculators, Mobile Phones and PCs of any kind are NOT allowed.

1. Define the following terms as used in Modeling and Simulation:
   a. Discrete system **[2 Mark]**
   b. Continuous system **[2 Mark]**
   c. Stochastic system **[2 Mark]**
   d. Deterministic system **[2 Mark]**
   e. Empirical Distribution **[2 Mark]**

2. (a) What is simulation? **[2Marks]**
   (b) Explain with flow chart, the steps involved in simulation study **[8 Marks]**

3. (a) Explain the following Kendall Notations of Queueing Systems:
   i. M/G/3/10 **[2 Marks]**
   ii. D/M/1/ 4 **[2 Marks]**
   iii. D/M/1/LILO/5/100 **[2 Marks]**
   (b) Mention any **four** situations when Kendall Models are appropriate tools **[4 Marks]**

4. A queuing model of grocery store has **one** checkout counter, which can serve **six** customers per hour. Assume customers arrive at this checkout counter at a rate of **ten** per hour, and each interval time is exponentially distributed.
   a. Describe the Kendall Notation of the Queuing System? **[2 Marks]**
   b. Calculate the performance of the Queuing System? **[6 Marks]**
   c. Describe **four** Service Discipline that may affect its performance **[8 Marks]**

5. (a) Using multiplicative congruential method (MCM), generate random numbers to a complete cycle? Given **A**=11, **M**=16, $\mathbf{X_0}$=7? **[8 marks]**
   (b) Explain the role of maximum density and maximum period for this MCM? **[6 marks]**

6. Consider a lifespan of a computer device used to detect faults in an aircraft is exponentially distributed and is given as $\lambda = {}^{1}/_{3}$ per Hour. Given that on average, there is **one** failure per **three** hours: -
   a. Calculate the Mean Life of the device? **[3 marks]**
   b. Calculate the probability that the device may last longer than its mean life **[5 marks]**
   c. Calculate the probability that the device may last between **2 to 3 Hours** **[5 marks]**
   d. Calculate the probability that the device may last for another hour given that it has been operating for **2.5** hours? **[7 marks]**

7. Consider a number of industrial injuries per working week in Castel factory is known to follow a Poisson Distribution with MEAN as $\lambda = {}^{1}/_{2}$.

a. Calculate the probability that in **one week period**, there will be:-

i. Less than 2 accidents **[7 marks]**

ii. More than 2 accidents **[7 marks]**

b. Calculate the probability that there will be no accidents **in three** weeks period **[6 marks]**

END OF QUESTION PAPER